Movie Recommendation System — Phase 2

Improving collaborative filtering with SVD++

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Predict unseen user–movie ratings

01

Phase 2

Establish the Phase 1 benchmark

02

SVD++

Introduce and tune SVD++

03

Model Ensemble

Combine it with User + Movie Bias

04

final_svdpp_bias_ensemble_submission

Generate the final Kaggle submission file.

RMSE

Root Mean Squared Error
LOWER IS BETTER

MovieLens at production scale

10M+
TRAINING RATINGS

5M+
TEST PAIRS

162,541
UNIQUE USERS

48,213
UNIQUE MOVIES

Ratings scale 0.5–5.0 • **Average 3.53**

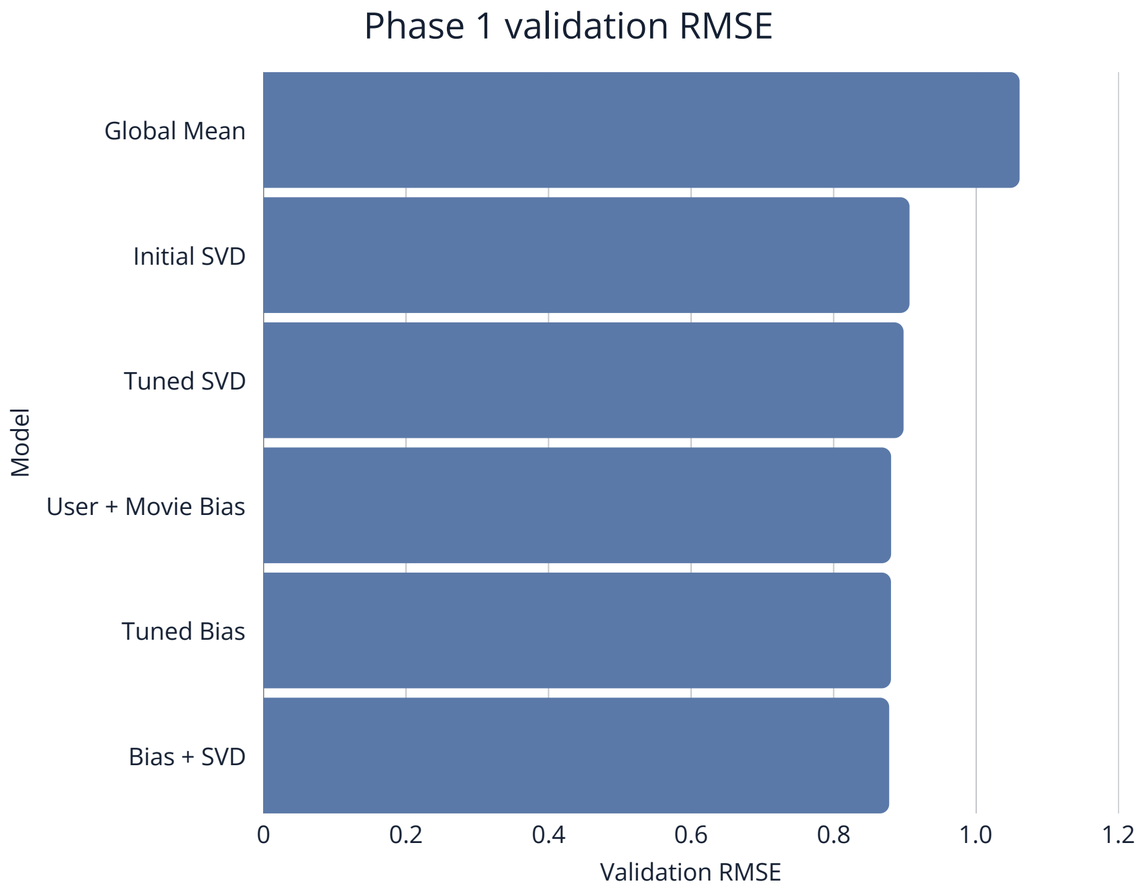
VALIDATION FRAMEWORK
80% training / 20% validation
Held constant across every experiment

0.877906 set the bar

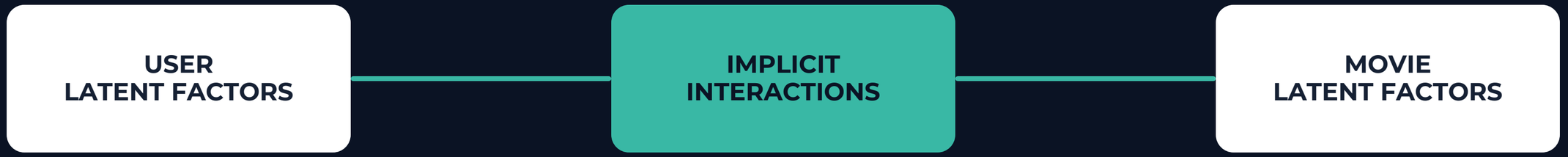
The Bias + SVD ensemble was the strongest Phase 1 validation result.

BENCHMARK

0.877906

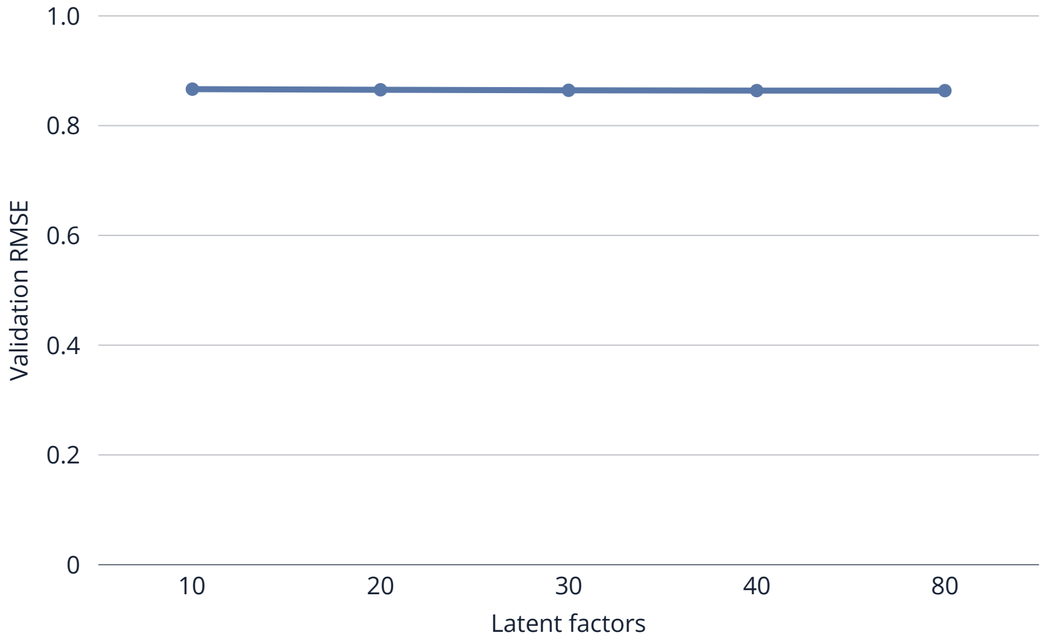


Latent factors, plus the signal of interaction

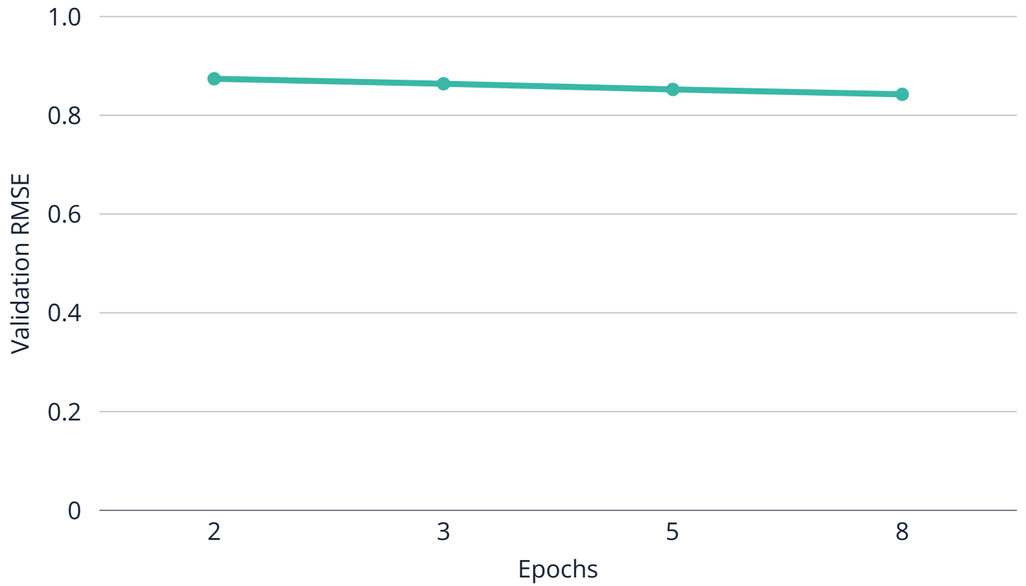


Can implicit feedback improve performance beyond Phase 1?

Factors experiment and epoch fixed at 3 : diminishing return



Epochs experiment and factors kept at 40 : consistent improvement



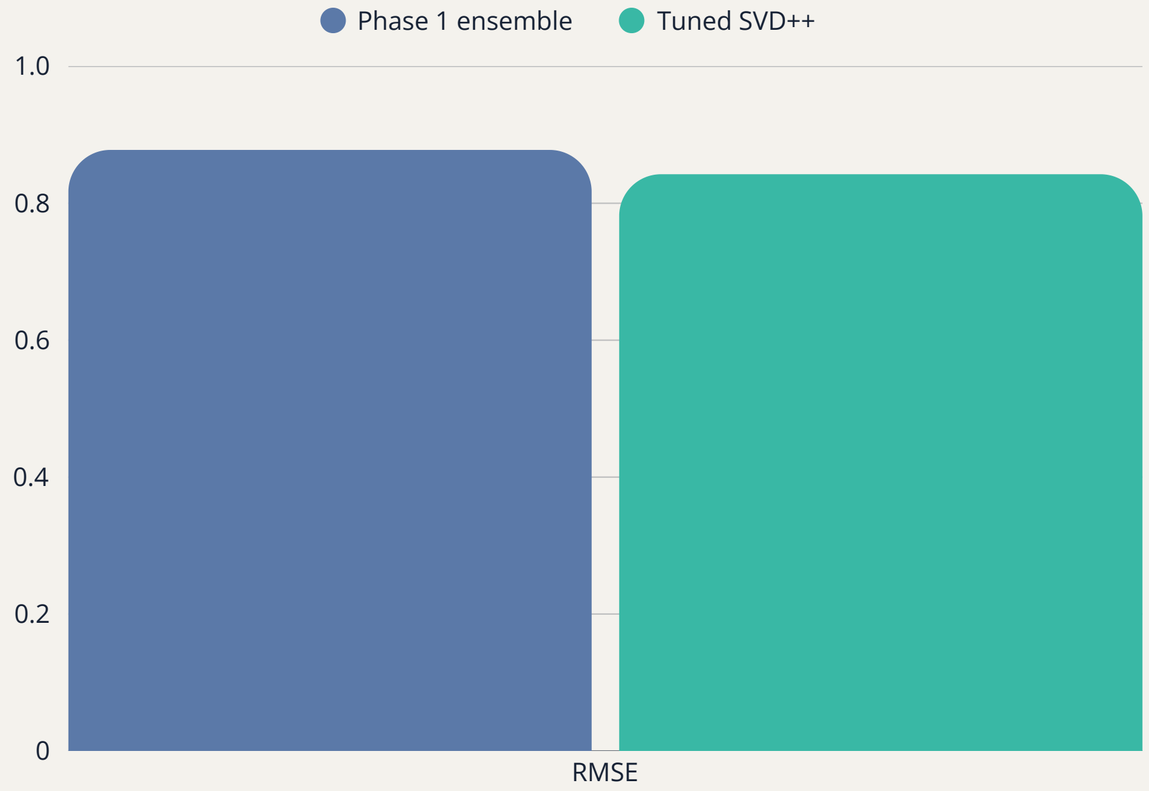
SELECTED 40 factors | 8 epochs | random_state 42 | RMSE 0.84239

A meaningful validation gain

−0.0355

RMSE reduction

SVD++ captured useful interaction information beyond the Phase 1 models.



Blend complementary signals



Final prediction = w_1 × SVD++ + w_2 × Bias

BEST ENSEMBLE

70 / 30 was the winning blend

70%
SVD++

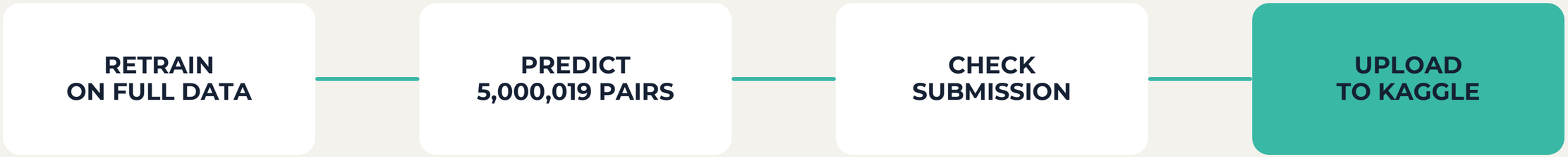
30%
USER + MOVIE BIAS

0.834601
VALIDATION RMSE

≈ 0.0078 improvement over SVD++ alone

From selected configuration to submission

SVD++ 40 factors · 8 epochs · seed 42 | Bias user reg 10 · movie reg 20 | Blend 70 / 30



Phase 2 moved the leaderboard

PHASE 1

~0.87

RANK 294

PHASE 2

~0.82

RANK 155

Competition results — distinct from 0.834601 validation RMSE

What the experiments taught me

KEY LESSONS

- Complexity is not a guarantee of quality
- Let validation guide tuning
- SVD++ adds useful interaction signal
- Diverse models can complement each other
- Compute cost shapes practical choices

NEXT EXPERIMENTS

- Hybrid collaborative + content models
- Genres and richer movie metadata
- Broader ensemble strategies
- More extensive hyperparameter search
- Advanced recommender architectures

Better models. Better blend. Better result.

0.834601
VALIDATION RMSE

~0.82
KAGGLE RMSE

155
LEADERBOARD RANK

Thank you / Questions?